

Task veridion

Big Data Engineer

Entity Resolution

Pasul 1:

-am incercat sa inteleg fisierul cu date,sa verific volumul de date

Initial m am gandit ca ,codul postal este un bun verificator de unicitate si am verificat duplicatele pe acesta coloana pentru a intelege mai bine.

Am realizat ca nu este suficient,deci am parcurs datele verificand mai intai duplicatele la nivel de coloana pe toate coloanele,initial am vazut valorile unice,apoi duplicatele, dar fara valorile nule

```
Coloana naics_2022_secondary_labels are duplicate:
naics_2022_secondary_labels
All Other Personal Services                      9
Lessors of Residential Buildings and Dwellings    7
Beauty Salons                                    6
Freight Transportation Arrangement                6
Telephone Apparatus Manufacturing                 5
Support Activities for Animal Production           4
Civic and Social Organizations                    4
Professional Organizations                        4
Recreational Goods Rental                         4
Sporting Goods Retailers                          4
All Other Industrial Machinery Manufacturing        3
Other Personal and Household Goods Repair and Maintenance 3
Other Services Related to Advertising              3
Child Care Services                              3
Marketing Consulting Services                     3
Other Personal Care Services                      3
Drinking Places (Alcoholic Beverages)             3
Packaging Machinery Manufacturing                 3
Offices of All Other Miscellaneous Health Practitioners 3
Testing Laboratories and Services                  2
Offices of Lawyers                               2
Motion Picture and Video Production                2
Ice Manufacturing                                2
Paint and Wallpaper Retailers                     2
General Automotive Repair                         2
Promoters of Performing Arts, Sports, and Similar Events without Facilities 2
All Other Traveler Accommodation                  2
```

```

Coloana phone_numbers are duplicate:
phone_numbers
+966920022097    35
+39800389822     26
+420800400450    25
+81568670314     23
+78124492121     23
..
+447512751439    2
+447780333047    2
+66978211111     2
+13015959000     2
+19106452600     2
Name: count, Length: 4236, dtype: int64

```

Pentru ca nu vedeam foarte bine datele in terminal,le am pus intr un fisier de tip json

```

1  {} date.json > ...
2  1
3  "company_name": {
4      "Fresh Burger": 30,
5      "Village Medical": 29,
6      "Recovera Využití zdrojů": 27,
7      "Söderberg & Partners": 27,
8      "Inter Cars": 26,
9      "Chatime": 26,
10     "Tomaticos": 24,
11     "China Merchants Securities Co. Ltd.": 22,
12     "Avangard": 21,
13     "Max Cash Advance": 20,
14     "Toys R Us": 20,
15     "Bike4you": 19,
16     "Betts Truck Parts & Service": 19,
17     "Arty Loon Show": 18,
18     "Ramsay Sante": 17,
19     "Wuai": 17,
20     "Advision": 16,
21     "Acustica Umbra Srl.": 16,
22     "Lojas Miamor": 16,
23     "Timpson Mobile Locksmiths": 15,
24     "Indus Towers": 15,
25     "SAHAR&VOSK": 15,
26     "Mexicali Cantina Grill": 15,
27     "Colutians TMB": 15

```

2.Pasul 2 a fost sa stabilesc ce coloane iau in considerare cand verific duplicitatea si sa imi propun sa folosesc aceste coloane pt a diferentia 2 companii doar daca au restul coloanelor la fel.

Si nu tin cont de coloanele: "company_type",

```
"lnk_year_founded",
"short_description",
"long_description",
"business_tags",
"business_model",
"product_type",
"naics_vertical",
"naics_2022_primary_code",
"naics_2022_primary_label",
"naics_2022_secondary_codes",
"naics_2022_secondary_labels",
"main_business_category",
"main_industry",
"main_sector",
"revenue",
"revenue_type",
"employee_count",
"employee_count_type",
"alexa_rank",
"sics_codified_industry",
"sics_codified_industry_code",
"sics_codified_subsector",
"sics_codified_subsector_code",
"sics_codified_sector",
"sics_codified_sector_code",
"sic_codes",
"sic_labels",
"isic_v4_codes",
"isic_v4_labels",
"nace_rev2_codes",
"nace_rev2_labels",
"website_tld",
"website_language_code",
"created_at",
"last_updated_at",
"website_number_of_pages",
"generated_description",
"generated_business_tags",
"status",
"domains",
"all_domains",
"inbound_links_count",
```

Dar o sa tin cont de astea :

- company_name
- company_legal_names
- company_commercial_names
- website_domain
- primary_email
- primary_phone
- linkedin_url
- main_country_code
- main_city
- main_postcode
- main_street
- main_latitude
- main_longitude
- year_founded
- naics_2022_primary_code
- main_industry
- main_sector

Pasul 3:

Problema este ca daca am cumva acelasi nume pentru 2 companii dar ele au restul campurilor diferite, companiile sunt diferite si dupa ideea de mai sus eu le as fi pus la fel pentru ca aveau primul camp identic. Asa ca o sa fac un fel de scor dupa 3 categorii, cele 2 de sus si plus unele secundare si in functie de unicitatea campului in categoria respectiv o sa ii dau un scor.

```
{
  "company_name": {
    "Fresh Burger": 30,
    "Village Medical": 29,
    "Recovera Využití zdrojů": 27,
    "Söderberg & Partners": 27,
    "Inter Cars": 26,
    "Chatime": 26,
    "Tomaticos": 24,
    "China Merchants Securities Co. Ltd.": 22,
    "Avangard": 21,
    "Max Cash Advance": 20,
    "Toys R Us": 20,
    "Bike4you": 19,
    "Betts Truck Parts & Service": 19,
    "Arty Loon Show": 18,
    "Ramsay Sante": 17,
    "Wuai": 17,
    "Advivision": 16,
    "Acustica Umbra Srl.": 16,
    "Lojas Miamor": 16,
    "Timpson Mobile Locksmiths": 15,
```

Spre exemplu Fresh Burger are 30 de duplicate deci compania o sa primesca un scor mai prost de unicitate in primul rand pt ca company_name ul este din categoria de coloane principale si pt ca sunt 30,daca erau 2 primea un scor mai bun

Pasul 4:

Dupa ce am stabilit un scor fiecarei companii ,caut sa le grupez .

Dupa ce am calculat scorul de unicitate pentru fiecare companie, am vrut sa adun la un loc toate inregistrarile care par sa fie aceeasi firma. Scopul a fost sa gasesc duplicatele si sa pastrez doar o varianta corecta pentru fiecare companie.

Am inceput prin a curata numele companiilor, adica am scos diferentele de scriere (litere mari, spatii, forme juridice ca SRL, LTD, INC) ca sa pot compara mai usor.

Apoi am folosit campuri care pot identifica direct o companie, cum ar fi domeniul site-ului, adresa de email sau numarul de telefon. Daca doua inregistrari aveau acelasi domeniu, le-am pus in acelasi grup.

Pentru restul companiilor am folosit o metoda numita fuzzy matching, care verifica cat de asemanatoare sunt doua denumiri chiar daca nu sunt identice. De exemplu, “Fresh Burger SRL” si “Fresh Burger” primesc un scor mare de asemanare si sunt puse in acelasi grup.

Am folosit un prag de asemanare de 92%. Daca doua companii aveau un scor peste acest prag, le consideram aceeaasi firma. Fiecare grup a primit un cod numit `fuzzy_group_id`.

La final, din fiecare grup am pastrat doar compania cu cel mai mare scor de unicitate, pentru ca e cea mai completa si relevanta.

Am salvat doua fisiere:

- `veridion_with_fuzzy_groups.parquet` – contine toate companiile, cu informatia despre grupul din care fac parte.
- `veridion_fuzzy_unique.parquet` – contine doar companiile unice, cate una pentru fiecare grup.